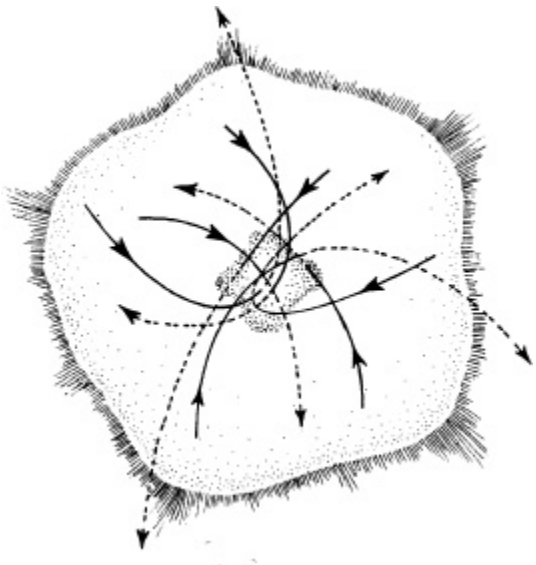
Not so for the singularity at the hole's center or in the Universe's final crunch, Khalatnikov and Lifshitz concluded. Their calculations seemed to show that tiny, random perturbations will grow large when the imploding matter attempts to create a singularity; they will grow so large, in fact, that they will prevent the singularity from forming. Presumably (though the calculations could not say for sure), the perturbations will halt the implosion and transform it into an explosion.

How could perturbations possibly reverse the implosion? The physical mechanism was not at all clear in the Khalatnikov–Lifshitz calculations. However, other calculations using Newton's laws of gravity, which are far easier than calculations using Einstein's laws, give hints. For example (see Figure 13.3), if gravity were weak enough inside an imploding star for Newton's laws to be accurate, and if the star's pressure were too small to be important, then small perturbations would cause different atoms to implode toward slightly different points near the star's center. Most of the imploding atoms would miss the center by some small amount and would swing around the center and fly back out, thereby converting the implosion into an explosion. It seemed conceivable that, even though Newton's laws of gravity fail inside a black hole, some mechanism analogous to this might convert the implosion into an explosion.



13.3 One mechanism for converting a star's implosion into an explosion, when gravity is weak enough that Newton's laws are accurate, and when internal pressure is weak enough to be unimportant. If the imploding star is slightly deformed ("perturbed"), its atoms implode toward slightly different points, swing around each other, and then fly back out.